Ivan Safonov

Python Backend Developer
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Summary

Python Backend Developer with 4+ years of experience building monolithic and event-driven microservices systems, including real-time analytics infrastructure for high-load media platforms (Python, Django, FastAPI, Kafka, ClickHouse). Experience integrating external AI APIs and ML models into production content and analytics pipelines, with a focus on system reliability, performance optimization, and measurable business impact.

Technical Skills

Programming Languages: Python, C++, SQL
Frameworks: Django, Flask, FastAPI
Databases: PostgreSQL, ClickHouse, Greenplum, Redis
Tools: Celery, Kafka, RabbitMQ, WebSocket, Pydantic, Apache Airflow, Apache Iceberg, scikit-learn
Infrastructure: Docker, Kubernetes, Nginx, Gunicorn, AWS, S3, Git, GitLab CI, CI/CD, Grafana, Prometheus
AI Tools: Claude Code, Codex

Experience

Nalitek Moscow, Russia
Software outsourcing company specializing in design, development, and support of high-load news portals and media platforms for enterprise clients.

Python Backend Developer Dec. 2023 – Present

- Reduced average reaction time to analytics events from 2 minutes to 0.2 seconds, which accelerated report generation by optimizing ClickHouse tables and processing a critical backlog of 40,000+ unprocessed messages.
- Designed a content creation automation module based on external LLMs and image-generation APIs (DRF, Celery, Kafka), which increased the average view count per content unit by 54.2%.
- Implemented an ML pipeline for celebrity recognition in photos and videos via KNN classification (face_recognition, scikit-learn) for a media library of 600,000+ files, reducing manual tagging by 70%.
- Root-caused a system-wide production failure across database, network, and infrastructure layers using Grafana metrics and controlled load testing in collaboration with frontend and DevOps engineers, restoring full content availability for all site visitors and boosting backend throughput 4x via a multithreaded Gunicorn migration.
- Developed an analytical event logging module (Kafka, ClickHouse, Celery, Postgres, Django signals) for tracking user interactions with content, replacing manual data exports with parameterized business reporting (XLSX/CSV) for editorial and management teams.

Contentline Moscow, Russia
IT contractor focused on state-commissioned development and maintenance of web platforms for agencies.

Python Backend Developer Aug. 2022 – Dec. 2023

- Built an automatic tag-recommendation engine using morphological analysis (pymorphy), eliminating manual tagging across the full editorial content pipeline.
- Optimized API endpoints for pages with large data volumes by debugging N+1 queries, which reduced average response time by 45.4%.
- Achieved up to 86% unit and integration test coverage across multiple projects, focusing on business-critical API endpoints, authentication flows, and async tasks, reducing bugs reaching production by 29%

Education

Central University Moscow, Russia
Master's in Machine Learning in Computer Science Sept. 2025 – Present
Bauman Moscow State Technical University Moscow, Russia
Bachelor's in Applied Informatics Sept. 2021 – Aug. 2025

Languages

English: B2 (Upper-Intermediate)
Russian: Native